9.10 Creating Histograms

Lesson Introduction

Benchmark	MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.	Learning Target	• I can create a histogram to represent data.

	Building On	Working Toward
Benchmark Coherence	MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.	MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.

Essential Questions	• What does the skew of the graph tell you? • Why can't you give the exact value of the median using the histogram? • How can you tell the difference between a gap, an outlier, and a cluster?
Lesson Narrative	Students will learn how to create histograms from a given set of data. They will describe the distribution's shape and features such as outliers, gaps, or clusters. Students will use the histogram to find the range and discuss the measures of center for the data.

Component Summary	9.10.1 Warm-Up (~5 min.)	Notice and wonder from a graphical display *(SE p. 97)*	9.10.3 Guided Practice (10-15 min.)	Complete a frequency table and histogram *(SE p. 100)*
	9.10.2 Guided Instruction (15-20 min.)	Work with a partner to calculate the range, interval width, minimum value, first interval, and table with frequencies *(SE p. 97)*	9.10.4 Your Turn! (5-8 min.)	Independently complete a frequency table and histogram for the given data *(SE p. 102)*
	9.10.5 Wrap-Up (~2 min.)	Reflect on the lesson and anything they are still confused by *(SE p. 102)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: • Frequency table <u>Additional Terms:</u> • Interval width • Frequency	• N/A	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may confuse left and right skew.
Students may misread the data, using the frequencies for the range instead of the x-axis.
Students may call every gap in a histogram an outlier instead of using the term for larger gaps.
Students may attempt to give the mean or median an exact value when they can usually only approximate the interval that contains the value.</td><td>Emphasize that the bars should touch in a histogram unless there is a gap or outlier.
Always use context when describing a distributions shape and features. You can use words like “unimodal” or “bimodal” if appropriate for the data.
The bin width is determined by width of the interval on the x-axis.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
9.10.1 Warm-Up engages the class in a Notice and Wonder routine. Encourage students to make connections between their notices and wonderings and those of their classmates to build investment in the content of the lesson.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
9.10.3 Guided Practice provides an opportunity for students to reflect on the mathematical thinking of others, particularly to recognize errors and explain how to fix them. Use this as an opportunity to engage the class in partner or whole-group discussion to explain their responses and articulate their thinking.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">9.10.2 Guided Instruction provides multiple entry points through the use of visuals and discussion-based learning. Consider ways in which kinesthetic entry points can be provided, such as online math manipulatives.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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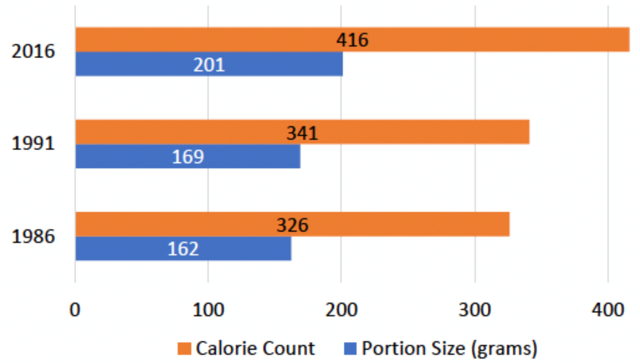
9.10.1 Warm-Up: Notice and Wonder

Component Overview: *In this Warm-Up, display the image for the class to see. Then have students participate in a Notice and Wonder routine.*

9.10.1 Warm-Up: Notice and Wonder

The portion size and calorie count of fast-food menu items have changed over time.



1. What are two things you notice about the graph?
   Answers will vary. Portion size is less than the calorie count. The graph shows two years that are five years apart and another that is 25 years apart.

2. What is one thing you wonder about the graph?
   Answers will vary. Why is calorie count increasing? Why is portion size increasing?

Notice and Wonder
Display the image for the class to see. Have students record what they notice about the image and what they wonder. This will engage all learners through two low-stakes questions. Bring the class back together to share their notices and wonderings. Consider recording on a T-Chart so students can make connections between their thoughts and those of their classmates.

9.10.2 Guided Instruction: Elements of Creating Histograms

Component Overview: *Students collaborate to determine how to use tables to answer questions about the data presented. Students first make observations of the data before they find the necessary components for their histogram.*

9.10.2 Guided Instruction: Elements of Creating Histograms

1. An earthworm farmer set up several containers of a certain species of earthworm so he could learn about their lengths. The farmer measured the lengths of 24 earthworms in one of the containers. Each length was measured in millimeters and recorded in the table.

6.5	11.2	18	19.6	20.1	23.8	23.5	25
25.7	26.2	27.5	27.5	28	29.9	32	33
41.4	42.3	48.8	52	54.4	59	60.6	77

Work with your partner to collect the pieces needed to create a histogram.

a. The range of the data is 70.5.

Determine the ***interval width*** by dividing the range and the number of intervals desired.

$$\frac{Range}{Number\ of\ Intervals} \approx Interval\ Width$$

b. With your partner, discuss three different numbers of intervals you could use with your partner, and how using a different number of intervals would affect the histogram.
Answers will vary. You could use $4, 6, or\ 8$. Using different intervals will affect the height of the histogram over certain intervals.

c. For this histogram, use six intervals. Divide the range by 6 and round up to the nearest whole number. This gives an ***interval width*** of 12.

This component is intentionally scaffolded to build on student understanding as the component progresses. Students begin by making observations about the data with a partner and then consider ways to represent the data in a frequency table and then a histogram. Ensure students understand how to create a frequency table using reasonable intervals, as this is imperative to being able to represent the data as a histogram.

Discuss how many intervals a histogram should have. The number of intervals depends on the data, but a typical histogram has five to ten intervals.

This component provides visual entry points through the use of tables and histograms. Auditory entry points are built in through partner and class discussion.

A ***frequency table*** is a table that shows how often each item, number, or range of numbers occurs in a set of data.

Use the steps below to create a frequency table to represent the data.

d. Complete each statement.

The minimum value is 6.5, so create the first interval that starts at 6 (or less).
To get the highest value in the interval, add the interval width of 12.

The first interval is $6-18$, the second interval is $18-30$.

e. With your partner, discuss if the value of 18 can be included in both intervals. Explain your conclusion.
No, each value can only be included in one interval. Otherwise, the frequencies would not be accurate.

f. Complete the **Length (mm)** column of the table shown to indicate the intervals.

Length (mm)	Frequency
$6-18$	2
$18-30$	12
$30-42$	3
$42-54$	3
$54-66$	3
$66-78$	1

g. With your partner, determine the frequency values and fill in the **Frequency** column of the table.
Shown in table.

Consider using visual aids, flash cards, and images to support the vocabulary and give the students time to grasp the meanings.

If you notice students struggling to determine which intervals to use, consider asking questions to further scaffold this component:

- *What do all intervals have in common?*
- *Will this always be true? Why or why not?*

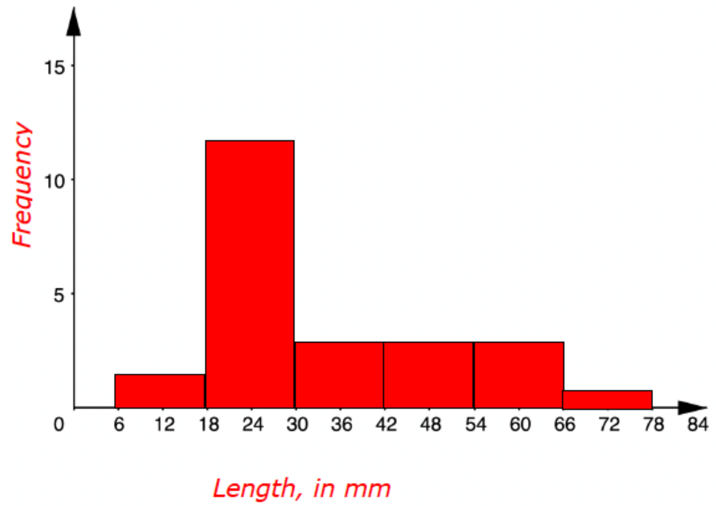
Specifically have students make connections to their responses to the first part of this question. Students who may not have made the connection to the importance of order might now better connect when given the context of a problem.

A data value of 18 should be included in the second interval, since the lower limit of an interval is included when finding frequency, but the upper limit is not.

9.10.2 Guided Instruction: Elements of Creating Histograms (continued)

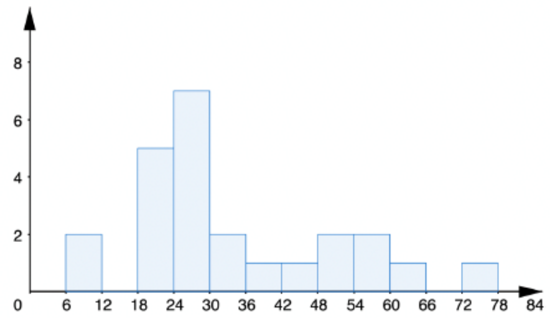
In a histogram, the bars touch. The variable measured is placed on the horizontal axis, and frequency is placed on the vertical axis.

h. Label the horizontal and vertical axis of the grid below. Then, use the grid to create the histogram to display the lengths of earthworms in millimeters.



The sum of the frequencies in the table will always equal the number of values in the data set. Students can also check their work by adding the frequency for each bar of their histogram.

2. The same data could be displayed in this histogram.



a. With your partner, discuss the differences you notice about this histogram from the one you created.
Answers will vary. The length of the interval width is 6; There are more bars in this histogram; There are haps in this histogram; The highest frequency is 7.

b. Explain which histogram better displays the data.
Answers will vary. Histogram #2, gives a better representation of the how many worms are within a specific measurement range and a more detailed view of the shape of the distribution.

c. Ask two classmates the reasons they listed for which histogram better displays the data. Record their choice and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.

Histogram Choice	My Summary of their Reason	Initials
	Answers will vary.	
	Answers will vary.	

d. Explain whether the data has the characteristics that will result in truthful data.
Answers will vary. Yes, the data has characteristics that result in truthful data. The measurement of the worm is accurate.

Notice and Wonder
Consider implementing a Notice and Wonder routine to draw connections between the two histograms.

- *What do you notice? What is the same?*
- *What is different?*
- *What does that make you wonder or think about this histogram?*
- *Will this always be true? Why or why not?*
- *How can we represent this graphically?*

9.10.3 Guided Practice: Creating Histograms

Component Overview: *In this Guided Practice, students create frequency tables and histograms based on provided data and then compare their interval widths with a partner. Students then analyze a provided work sample to determine if the data presented is represented correctly in a frequency table and histogram.*

9.10.3 Guided Practice: Creating Histograms

1. The data shows the number of Snapchats sent by 12 seventh-grade students in one day.

$$99, 97, 94, 88, 84, 81, 80, 77, 71, 63, 60, 54$$

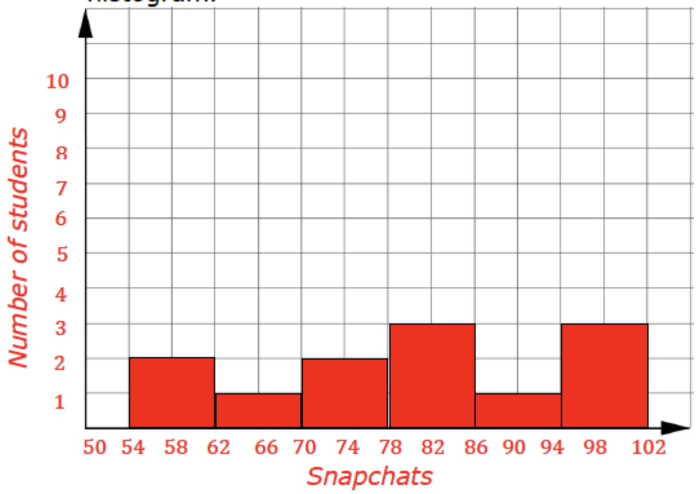
Complete the following to create a frequency table and a histogram to represent the number of Snapchats sent by seventh-grade students in one day.

a. How many intervals are appropriate to use for this histogram?
Answers will vary. 5

b. List three possible interval widths.
Answers will vary. 5, 6, 8

c. Using one of the possible interval widths, create the frequency table and histogram of the data. Make sure to label both the horizontal and vertical axis of the histogram.

Snapchats	Frequency
54 – 62	2
62 – 70	1
70 – 78	2
78 – 86	3
86 – 94	1
94 – 102	3



Consider building in partner discussion throughout this component to provide auditory entry points for learners. For students who may be struggling readers, it may be helpful to read the questions out loud. Consider pulling a small group to do this and allowing other students to work in pairs or small groups independently.

If you notice students struggling to create the histograms, ask scaffolding questions like:

- *What intervals did you use in your frequency table?*
- *How can you represent these intervals on the x-axis? What scale should you use?*

9.10.3 Guided Practice: Creating Histograms (continued)

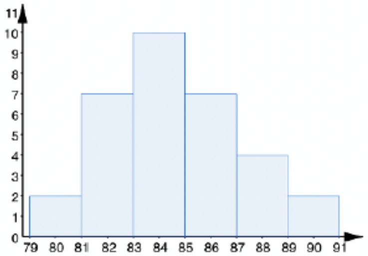
d. Find a classmate who used a different interval width and compare your histograms. Summarize your discussion on what you notice about the two different graphs.
Answers will vary. They used larger interval widths, resulting in less number of intervals than I had.

2. Raisa used data she collected about the resting heart rates, in beats per minute (bpm), of 22 classmates to create a histogram. The heart rates are listed below.

89 87 85 84 90 79 83 85 86 88 84 81 88 85 83
83 86 82 83 86 82 84

She created a frequency table and used it to make a histogram. The frequency table and histogram Raisa created are shown below. Unfortunately, Raisa noticed the data she collected was not accurately displayed.

Heart Rate (bpm)	Frequency
79 – 81	2
81 – 83	7
83 – 85	10
85 – 87	7
87 – 89	4
89 – 91	2



Heart Rate (bpm)

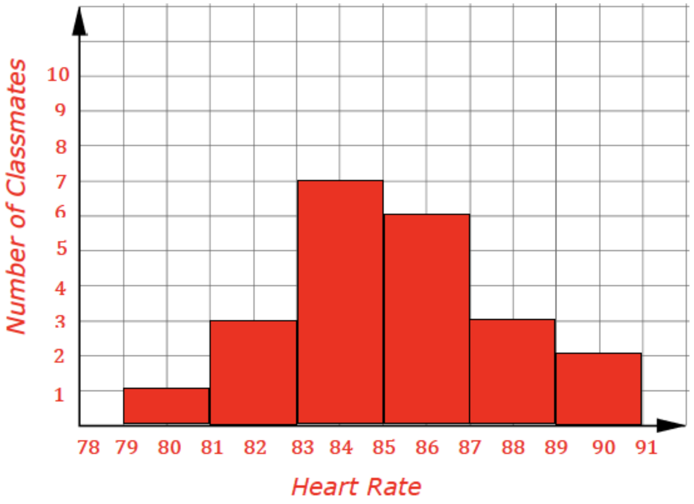
a. Explain what Raisa did wrong.
Raisa counted the cut scores twice.

MA.K12.MTR.4.1: Discussion and Reflection
Question 2 provides an opportunity for students to reflect on the mathematical thinking of others, particularly to recognize errors and explain how to fix them. Use this as an opportunity to engage the class in partner or whole-group discussion to explain their responses and articulate their thinking.

9.10.3 Guided Practice: Creating Histograms (continued)

b. Correct Raisa's mistake and create a frequency table and histogram that correctly displays the data she collected.

Heart Rate (bpm)	Frequency
79 – 81	1
81 – 83	3
83 – 85	7
85 – 87	6
87 – 89	3
89 – 91	2



Monitor for students who make similar mistakes to Raisa. Consider the following prompting questions:

- *Why are numbers like* 81 *repeated in the table of values?*
- *Which group of frequencies would a number like* 81 *count for? Why?*
- *What does that look like on your histogram?*

9.10.4 Your Turn!

Component Overview: *Students work independently to create a frequency table and histogram to represent the number of chips in a bag.*

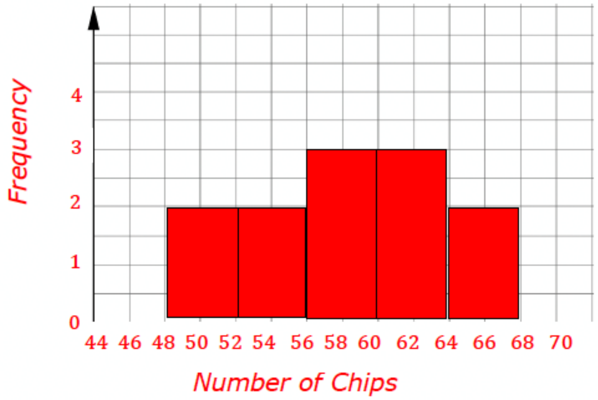
9.10.4 Your Turn!

1. Sergio noticed that a bag of chips advertised that the serving size was 15 chips and that the bag contained approximately 4 servings. He counted the number of chips in a sample of 12 bags and recorded the total number of chips below.

$$48, 51, 52, 55, 56, 58, 59, 60, 61, 62, 65, 67$$

Create a frequency table and histogram for the number of chips in a bag. Answers may vary.

Number of Chips	Frequency
48 – 52	2
52 – 56	2
56 – 60	3
60 – 64	3
64 – 68	2



Students may struggle to determine an interval width that makes sense with the data. Ask guiding questions to help students determine their interval width:

- *What is the range of the data? How can you use this to help determine the interval width?*
- *How many widths are appropriate for this data set? Why?*

9.10.5 Wrap-Up: Reflection

Component Overview: *Students reflect on the most confusing part of the day's lesson.*

9.10.5 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on creating histograms?

Answers will vary.

Consider providing the learning targets as sentence stems for students who are unsure how to capture muddy points from the lesson.